

SKILL 22

```
#include <stdio.h>
#include <stdlib.h>

int add(int a, int b) {
    return a + b;
}

int main() {
    int *numbers;

    printf("=== Task 22 Program ===\n");

    numbers = malloc(5 * sizeof(int));

    if (numbers == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++) {
        numbers[i] = i + 1;
    }

    printf("Stored values: ");

    for (int i = 0; i < 5; i++) {
        printf("%d ", numbers[i]);
    }

    printf("\n");

    printf("Addition Test: 10 + 20 = %d\n", add(10, 20));

    free(numbers);

    printf("Memory released successfully\n");
    printf("Program completed successfully\n");

    return 0;
}
```

```
nishanth@LAPTOP-6Q3ON07H:~/task22$ nano task22.c
nishanth@LAPTOP-6Q3ON07H:~/task22$ gcc -g task22.c -o task22
nishanth@LAPTOP-6Q3ON07H:~/task22$ ./task22

=== Task 22 Program ===
Stored values: 1 2 3 4 5
Addition Test: 10 + 20 = 30
Memory released successfully
Program completed successfully
```

```
nishanth@LAPTOP-6Q3ON07H:~/task22$ sudo apt update
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [550 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [716 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [167 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.0 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [315 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [101 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [486 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [129 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [97.7 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.9 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3276 B]
Get:17 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [191 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [60.4 kB]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.6 kB]
Get:22 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [445 kB]
Get:23 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [89.7 kB]
Get:24 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [10.8 kB]
Get:25 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2844 B]
Fetched 4192 kB in 5s (774 kB/s)
157 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
157 packages can be upgraded. Run 'apt list --upgradable' to see them.
nishanth@LAPTOP-6Q3ON07H:~/task22$ valgrind --leak-check=full ./task22
==692== Memcheck, a memory error detector
==692== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==692== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==692== Command: ./task22
==692==
=== Task 22 Program ===
Stored values: 1 2 3 4 5
Addition Test: 10 + 20 = 30
Memory released successfully
Program completed successfully
==692==
==692== HEAP SUMMARY:
==692==   in use at exit: 0 bytes in 0 blocks
==692==   total heap usage: 2 allocs, 2 frees, 1,044 bytes allocated
==692==
==692== All heap blocks were freed -- no leaks are possible
==692==
==692== For lists of detected and suppressed errors, rerun with: -s
==692== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

```
#!/bin/bash_

echo "=== Task 22 Automated Testing ==="

echo "Compiling program..."

gcc -g task22.c -o task22

if [ $? -ne 0 ]; then
    echo "Compilation FAILED"
    exit 1
fi

echo "Compilation PASSED"

echo
echo "=== Running Program ==="

./task22 > actual.txt

cat actual.txt

echo
echo "=== Creating Expected Output ==="

cat > expected.txt << EOF
=== Task 22 Program ===
Stored values: 1 2 3 4 5
Addition Test: 10 + 20 = 30
Memory released successfully
Program completed successfully
EOF
```

```
echo
echo "=== Comparing Results ==="

if diff expected.txt actual.txt > /dev/null; then
    echo "TEST PASSED"
    result="PASS"
else
    echo "TEST FAILED"
    result="FAIL"
fi

echo
echo "=== Running Valgrind ==="

valgrind --leak-check=full \
    --error-exitcode=1 \
    ./task22 > /dev/null 2> valgrind_report.txt

if [ $? -eq 0 ]; then
    echo "VALGRIND TEST PASSED"
    memory_result="PASS"
else
    echo "VALGRIND TEST FAILED"
    memory_result="FAIL"
fi

echo
echo "=== Generating Report ==="

cat > test_report.txt << EOF
TASK 22 TEST REPORT

Program Test: $result
Memory Test: $memory_result

Files checked:
- task22.c
- actual.txt
- expected.txt
- valgrind_report.txt

Testing completed.
EOF
```

```
cat test_report.txt

echo
echo "=== Task 22 Complete ==="
```

```
nishanth@LAPTOP-6Q3ON07H:~/task22$ chmod +x test.sh
nishanth@LAPTOP-6Q3ON07H:~/task22$ ./test.sh
=== Task 22 Automated Testing ===
Compiling program...
Compilation PASSED

=== Running Program ===
=== Task 22 Program ===
Stored values: 1 2 3 4 5
Addition Test: 10 + 20 = 30
Memory released successfully
Program completed successfully

=== Creating Expected Output ===

=== Comparing Results ===
TEST FAILED

=== Running Valgrind ===
VALGRIND TEST PASSED

=== Generating Report ===
TASK 22 TEST REPORT

Program Test: FAIL
Memory Test: PASS

Files checked:
- task22.c
- actual.txt
- expected.txt
- valgrind_report.txt

Testing completed.
```

```
=== Task 22 Complete ===
nishanth@LAPTOP-6Q3ON07H:~/task22$ cat test_report.txt
TASK 22 TEST REPORT

Program Test: FAIL
Memory Test: PASS

Files checked:
- task22.c
- actual.txt
- expected.txt
- valgrind_report.txt

Testing completed.
```